



Protein Embeddings and Local Alignments

Malec et al. (2025)

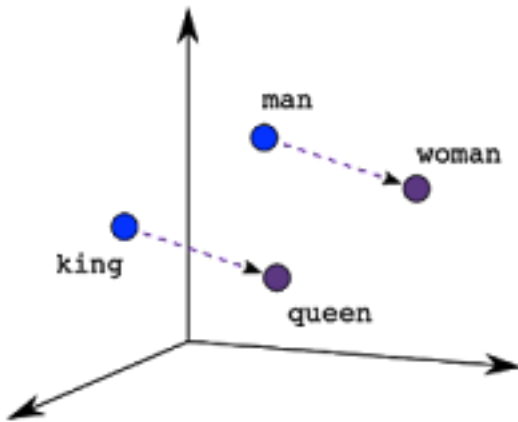
Serena Chan

Current Methods

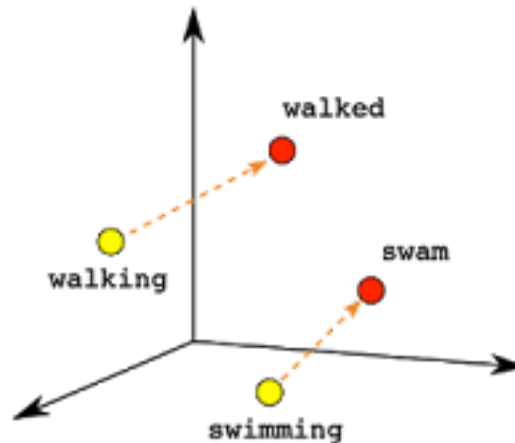
- Local alignments are crucial for discovering regions of similarity within different sequences
- BLOSUM matrices is the standard scoring method, however accurate alignment of distantly related sequences remains a challenge

Vector Embeddings

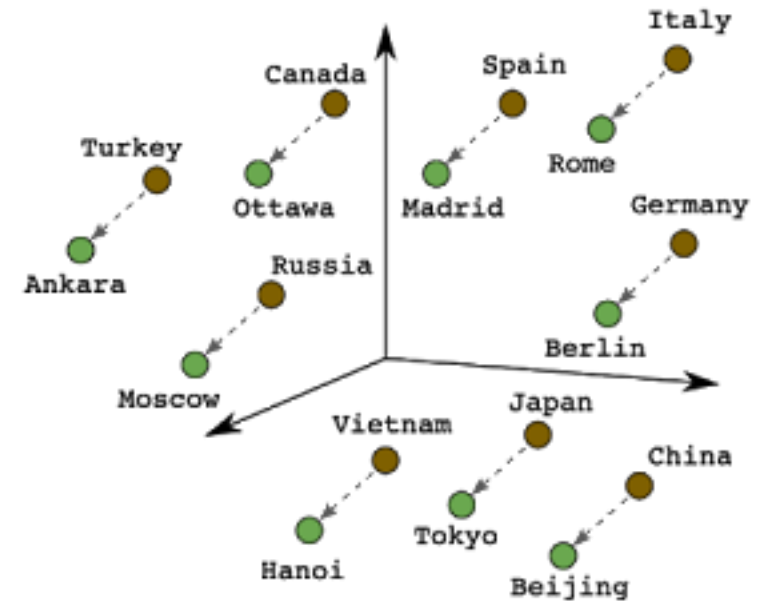
- Embeddings consist of numerical vectors encoded with semantic meaning
- Originated from Natural Language Processing (NLP)



Male-Female



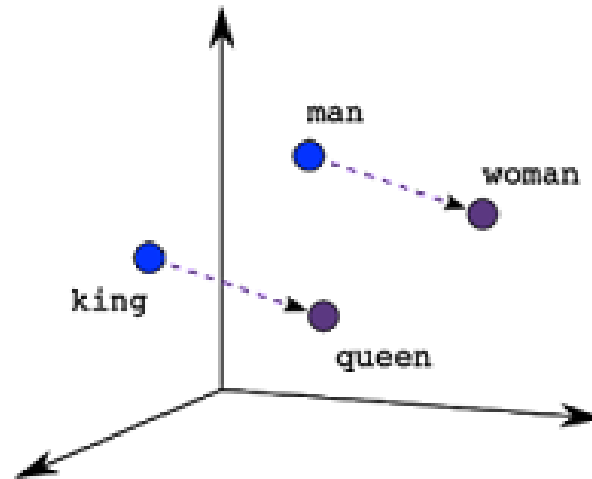
Verb Tense



Country-Capital

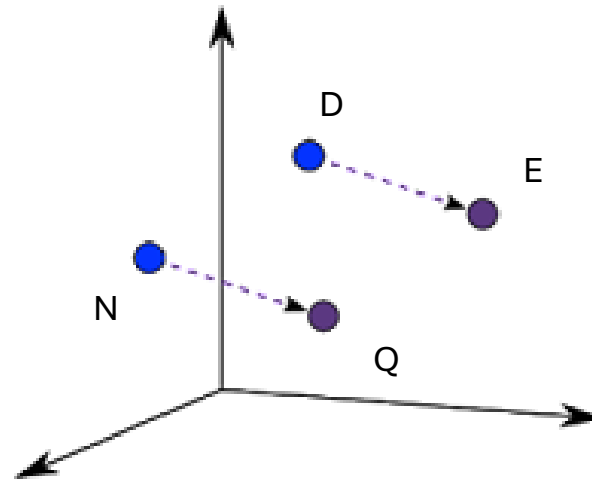
Protein Embeddings

- Rather than containing words, we embed protein information
- The similarity between two residues can then be calculated using the cosine similarity of their embedding vectors
- These do not replace alignments; rather improve the quality of alignments



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Objective #1

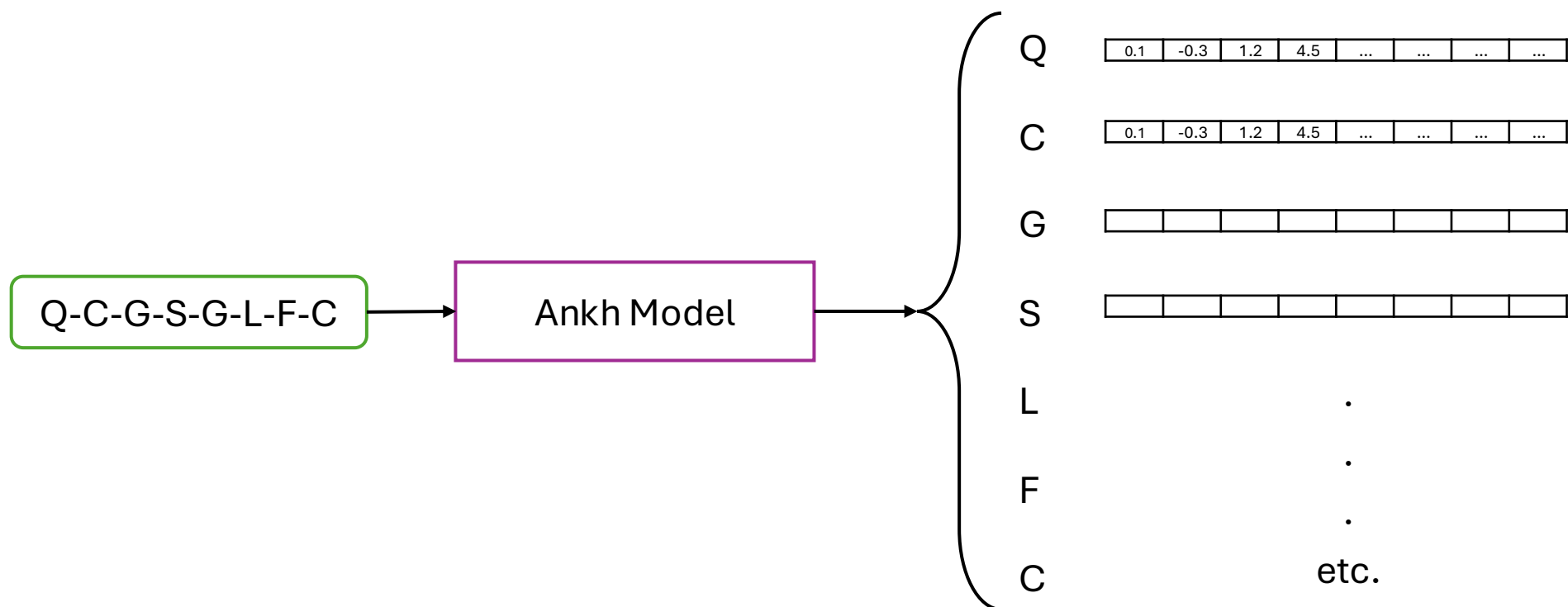
Develop a superior alignment algorithm to BLOSUM to increase accuracy in distant alignment

Alignment Computation

1. Generate vector embeddings
2. Calculate E-score between every possible pair of residues between sequences
3. Run Smith-Waterman algorithm using:
 - Match/Mismatch: E-scores
 - Indels: affine gaps (Open = -1, Ext = -0.1)
4. Evaluate alignment against reference alignment

1. Generate Embeddings

- Evaluated 7 protein language models by difference between the computed and reference alignment within the same domain



2. Calculate E-Score

- E-score: cosine similarity between two embeddings
 - If the vectors point in the same direction, score is higher and vice versa.

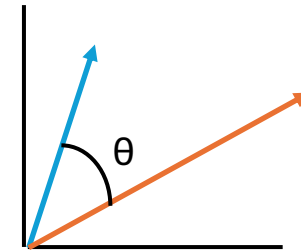
$$E\text{-score}(a_1, a_2) = \frac{E(a_1) \cdot E(a_2)}{\|E(a_1)\| \|E(a_2)\|}$$

Sequence A

0.1	-0.3	1.2	4.5	...	...	...	...
-----	------	-----	-----	-----	-----	-----	-----

Sequence B

0.1	-0.3	1.2	4.5	...	...	...	...
-----	------	-----	-----	-----	-----	-----	-----



- Rather than looking up the score in a BLOSUM table, the matrix is populated with the E-scores.

3. Smith-Waterman DP

- As usual... but with E-score scoring matrix

		G	W	A	A	P	H	E	T	A
		0	0	0	0	0	0	0	0	0
T		0	0	3	3	1	0	0	3	1
W		0	3	1	1	6	4	2	1	0
A		0	1	1	4	4	3	1	5	3
A		0	0	0	4	7	5	3	4	2
H		0	0	0	2	6	10	8	6	7
E		0	0	0	0	4	8	13	11	9
G		0	3	3	1	3	6	11	10	8
G		0	3	6	4	2	6	9	8	7

4. Evaluate Computed Alignment

- A successful local alignment method must correctly address:

Localization

P K E L M V P L A

Q C G S G L F C

A N C L K K S H S

Quality

Q C G S G L F C

A B C D E F G H

Q C S S - L F C

4. Evaluate Localization of Alignment

- A computed alignment can differ from reference by being:
 - Shorter (under-extension)
 - Misses part of reference
 - Longer (over-extension)
 - Extends outside of reference
 - Offset
 - Combination of shorter and longer
- This results in an inaccurate representation of the target region

Objective #2

Establish a comprehensive alignment evaluation framework

4. Evaluate Localization of Alignment

- **Solution:** force the computed result to align the full reference sequence by constructing the final alignment in 3 segments:

$$(s''_1, s''_2) = \text{WORST}(\ell_1, \ell_2)(o'_1, o'_2)\text{WORST}(r_1, r_2)$$

Ex: p_1: PKELMVPLAEMECQQC**QCGSGLFC**ANCLKKSHS
p_2: LAEYECRDC**SLQAGSGLAFC**VCLQHWNGE

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Reference

s_1: QCGSGLFC

s_2: SLQAGSGLAFC

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Reference

s_1: QCGSGLFC

s_2: SLQAGSGLAFC

s'_1: --QCGSGL--FC

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Ex: p_1: PKELMVPLAEMECQQCQCGSGLFCANCLKKSHS
p_2: LAEYECRDCSLQAGSGLAFCVCLQHWNGE

Reference	Computed
s_1: QCGSGLFC	t_1: GSGLFCANCL
s_2: SLQAGSGLAFC	t_2: GLAFCVCL
s'_1: --QCGSGL--FC	
s'_2: SLQAGSGLAFC	

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- Solution: force the computed result to align the full reference sequence by constructing the final alignment in 3 segments:

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Ex: p_1: PKELMVPLAEMECQQCQCGSGLFCANCLKKSHS
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Reference	Computed
s_1: QCGSGLFC	t_1: GSGLFCANCL
s_2: SLQAGSGLAFC	t_2: GLAFCVCL
s'_1: --QCGSGL--FC	t'_1: GSGLF--CANCL
s'_2: SLQAGSGLAFC	t'_2: G--LAFCV--CL

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Ex: p_1: PKELMVPLAEMECQQCQC GSGLFCANCLKKSHS
p_2: LAEYECRDCSLQAGSGLAFCVCLQHWNGE

Reference	Computed
s_1: QCGSGLFC	t_1: GSGLFCANCL
s_2: SLQAGSGLAFC	t_2: GLAFCVCL
s'_1: --QCGSGL--FC	t'_1: GSGLF--CANCL
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Ex: p_1: PKELMVPLAEMECQQC**QC****GSGLFC****ANCL**KKSHS
p_2: LAEYECRDC**SLQAGSGLAFC****VCL**LQHWNGE

Reference	Computed	Overlap
s_1: QC GSGLFC	t_1: GSGLFC ANCL	o_1: GSGLFC
s_2: SLQAGSGLAFC	t_2: GLAFC VCL	o_2: GLAFC
s'_1: -- QC GSGL -- FC	t'_1: GSGLF -- CANCL	
s'_2: SLQAGSGLAFC	t'_2: G -- LAFCV -- CL	

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Ex: p_1: PKELMVPLAEMECQQC**QC****GSGLFC****ANCL**KKSHS
p_2: LAEYECRDC**SLQAGSGLAFC****VCL**QHWNGE

Reference	Computed	Overlap
s_1: QC GSGLFC	t_1: GSGLFC ANCL	o_1: GSGLFC
s_2: SLQAGSGLAFC	t_2: GLAFC VCL	o_2: GLAFC
s'_1: -- QC GSGL -- FC	t'_1: GSGLF -- CANCL	o'_1: GSGL -- FC
s'_2: SLQAGSGLAFC	t'_2: G -- LAFCV -- CL	o'_2: G -- LAFC

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$$(s''_1, s''_2) = \text{WORST}(\ell_1, \ell_2)(o'_1, o'_2)\text{WORST}(r_1, r_2)$$

Ex: p_1: PKELMVPLAEMECQQCQC GSGLFCANCLKKSHS
p_2: LAEYECRDC SLQAGSGLAFCVCLQHWNGE

Reference	Computed	Overlap	Not Computed
s_1: QC GSGLFC	t_1: GSGLFCANCL	o_1: GSGLFC	L_1: QC
s_2: SLQAGSGLAFC	t_2: GLAFCVCL	o_2: GLAFC	L_2: SLQAGS
s'_1: --QC GSGL--FC	t'_1: GSGLF--CANCL	o'_1: GSGL--FC	
s'_2: SLQAGSGLAFC	t'_2: G--LAFCV--CL	o'_2: G--LAFC	

4. Evaluate Localization of Alignment

- Solution: force the computed result to align the full reference sequence by constructing the final alignment in 3 segments:

$$(s''_1, s''_2) = \text{WORST}(\ell_1, \ell_2)(o'_1, o'_2)\text{WORST}(r_1, r_2)$$

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Reference	Computed	Overlap	Not Computed
s_1: QCGSGLFC	t_1: GSGLFCANCL	o_1: GSGLFC	L_1: QC
s_2: SLQAGSGLAFC	t_2: GLAFCVCL	o_2: GLAFC	L_2: SLQAGS
s'_1: --QCGSGL--FC	t'_1: GSGLF--CANCL	o'_1: GSGL--FC	r_1: ε
s'_2: SLQAGSGLAFC	t'_2: G--LAFCV--CL	o'_2: G--LAFC	r_2: ε

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- Solution: force the computed result to align the full reference sequence by constructing the final alignment in 3 segments:

$$(s''_1, s''_2) = \text{WORST}(\ell_1, \ell_2)(o'_1, o'_2)\text{WORST}(r_1, r_2)$$

Ex: p_1: PKELMVPLAEMECQQC**QC****GSGLFC****ANCL**KKSHS
p_2: LAEYECRDC**SLQAGS****GLAFC****VCL**LQHWNGE

Final s''_1: **QC**-----**GSGL-FC**
s''_2: --**SLQAGSG**--**LAFC**

Reference	Computed	Overlap	Not Computed
s_1: QC GSGLFC	t_1: GSGLFC ANCL	o_1: GSGLFC	l_1: QC
s_2: SLQAGS GLAFC	t_2: GLAFC VCL	o_2: GLAFC	l_2: SLQAGS
s'_1: -- QC GSGL-FC	t'_1: GSGLF- CANCL	o'_1: GSGL-FC	r_1: ϵ
s'_2: SLQAGS GLAFC	t'_2: G -- LAFCV-CL	o'_2: G -- LAFC	r_2: ϵ

4. Evaluate Quality of Alignment

- The quality was measured using 5 distances:
 - d_ssp: Sum of Pairs distance
 - d_seq: Sequence Gap distance
 - d_pos: Position Gap distance
 - **d_d: Relative Displacement Distance**

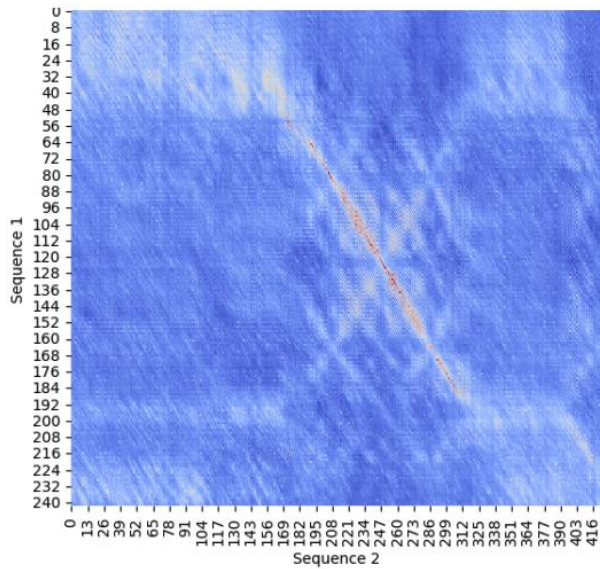
$$d_d(A_1, A_2) = \frac{1}{mn(m+n)} \sum_{i=1}^n \sum_{j=1}^m |d_{A_1}(i, j) - d_{A_2}(i, j)|.$$

- **d_cc: Closest Context Distance**

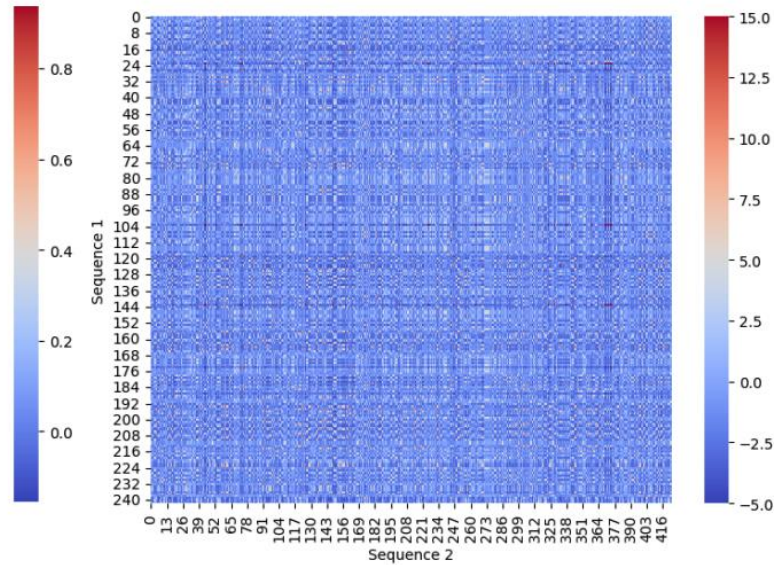
$$d_{cc}(A_1, A_2) = \frac{1}{4mn} \sum_{\ell=1}^2 \left(\sum_{i=1}^n d_{P,i,\ell} + \sum_{j=1}^m d_{Q,j,\ell} \right).$$

Results

- E-score vs. BLOSUM
 - Across all natural datasets/domains, BLOSUM 45, 50, 62, 80, and 90 each performed worse than using the E-score



E-Score



BLOSUM

Complexity

- Time

- Computation: $O(n*m*d)$
 - Embeddings: $O(n*m)$
 - E-score matrix: $O(n*m*d)$
 - DP: $O(n*m)$
- Evaluation: $O(n+m)$
 - Padding: $O(n+m)$
 - Distance metrics: $O(n+m)$

- Space

- Embeddings: $O(d(n+m))$
- E-score matrix: $O(n*m)$
- DP matrix: $O(n*m)$

Limitations

- Lower performance on artificial sequence datasets
- Computationally more expensive than a simple BLOSUM lookup
- Padding introduced to solve an issue that E-score-based alignments created; but there is no method that eliminates the overhangs in general
- Cannot be combined with other embedding methods

References

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Thank you for your
time!

